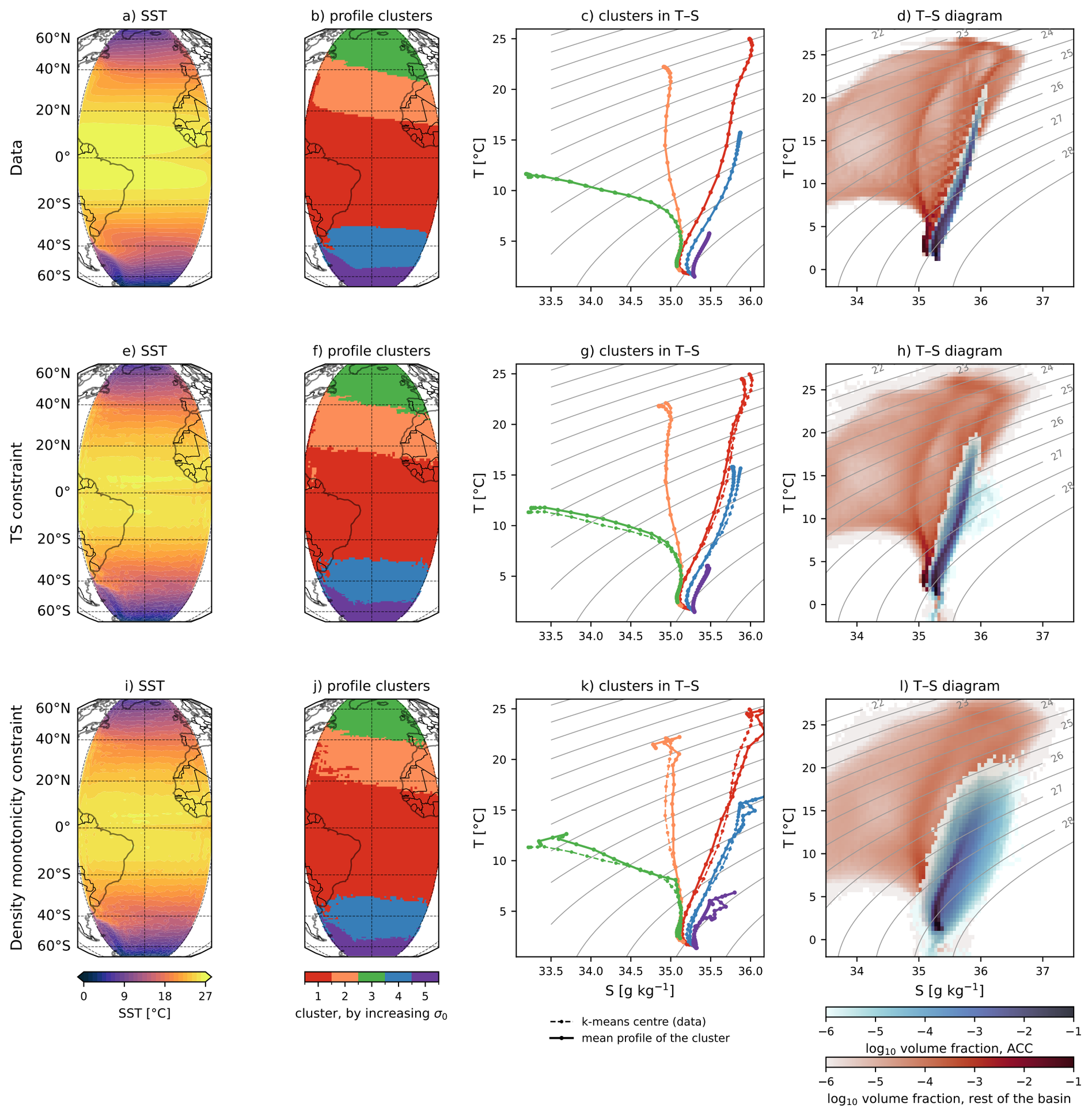

a) SST
b) profile clusters
c) clusters in T–S
d) T–S diagram
e) SST
f) profile clusters
g) clusters in T–S
h) T–S diagram
i) SST
j) profile clusters
k) clusters in T–S
l) T–S diagram
Data
TS constraint
Density monotonicity constraint
60°N
40°N
20°N
0°
20°S
40°S
60°S
T [°C]
S [g kg^{-1}]
0 9 18 27
SST [°C]
1 2 3 4 5
cluster, by increasing σ_0
k-means centre (data)
mean profile of the cluster
−6 −5 −4 −3 −2 −1
log$_{10}$ volume fraction, ACC
log$_{10}$ volume fraction, rest of the basin